



“QNU Labs and India will be the go-to place for quantum cybersecurity”

We live in a digital world that's heavily dependent on the internet, and yet, many of our cybersecurity methods are pretty juvenile. QNu Labs, pronounced as Q-New, is India's only quantum-tech cybersecurity company, and it is trying to lead the quantum mission for India. **Sunil Gupta, co-founder & CEO of QNU Labs**, explains the need to reimagine cybersecurity in an interaction with **EFY**

» **Sunil Gupta**
co-founder & CEO, QNU Labs

Q Why do you feel there is a need to reimagine cybersecurity?

A. Let me take you back to 2016. About five-and-a-half years ago, there were several things happening in India and also globally. In 2016, demonetisation happened and India decided to go digital—from a cash based economy to a cashless economy. PhonePe, UPI, etc, came up. Everybody in the country has started doing things using digital platforms. India is becoming a truly digital economy,

except for a few cash based activities. Healthcare, social security, and banking, all happen digitally.

First of all, today's encryption is based upon a complex mathematical problem that can't be broken by current classical computers and supercomputers but hackers will be able to break the encryption using scalable quantum computers. The second thing is that encryption keys today are generated once in a while. They should be generated more frequently and should be random. You will be surprised to

find out that large enterprises typically have about 85,000 keys. Most of these keys do not change for two to three years! Now imagine if a key is stolen by somebody and the person decrypts the data and steals it.

Whenever a breach happens, people are worried about data loss, but they do not know that they also lose the encryption keys. When people hack into a server, they not only take away the data but also the encryption keys—so that they can steal data in the future as well.